

Optimization 1

ISE 5406

Lecture 7: Duality Theory

Dr. Robert Hildebrand

Grado Department of Industrial & Systems Engineering
Virginia Tech

Where We Are in the Course

Last Time (Lecture 6): Constrained Optimization and KKT Conditions

- Lagrangian function and Lagrange multipliers
- Karush-Kuhn-Tucker (KKT) necessary and sufficient conditions
- Stationarity, primal/dual feasibility, complementary slackness
- Constraint qualifications (Slater's condition)

Today (Lecture 7): Duality Theory

- The Lagrangian dual problem and dual function
- Weak duality theorem: $d^* \leq p^*$
- Strong duality and when it holds (Slater's condition)
- Economic interpretation: dual variables as shadow prices
- Application to fractional knapsack problem

Next Time (Lecture 8): Gradient Descent and Convergence

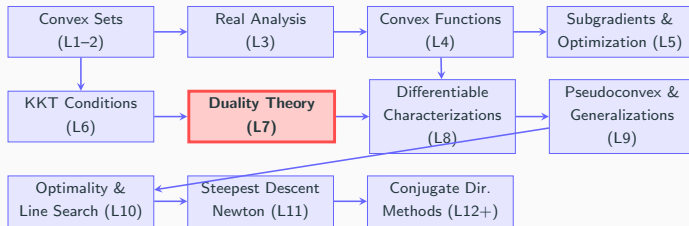
- Gradient descent algorithm and convergence analysis
- Convergence rates for convex and strongly convex functions
- Step size selection strategies

Coming Soon (Lecture 9-10): Projected Gradient and Proximal Methods

- Projected gradient descent for constrained optimization
- Proximal operators and proximal gradient methods
- Applications to large-scale optimization

Big Picture: Duality bridges theory and computation. It provides optimality certificates (weak duality), characterizes optimal solutions (strong duality), and guides algorithm design by revealing problem structure through dual variables.

Big Picture: Course Roadmap



Why Duality Matters

- Provides lower bounds on optimal value (weak duality)
- Enables optimality verification without solving problem (certificate of optimality)
- Reveals economic meaning of constraints through shadow prices
- Foundation for many efficient algorithms (cutting planes, column generation, etc.)

Reading for This Lecture

Textbook References

- Sections 5.1–5.2 of Boyd and Vandenberghe (Convex Optimization)
- Sections 6.1–6.2 of Bazaraa, Sherali, and Shetty (Nonlinear Programming)
- Key concepts: Lagrangian dual function, weak/strong duality, Slater's condition

Supplemental Video — KKT Review

For a great visual explanation of KKT conditions (reviewed in Lecture 6), see:

<https://www.youtube.com/watch?v=uh1Dk68cfWs>

Covers: geometric intuition for Lagrange multipliers, inequality constraints, stationarity, complementary slackness, and constraint qualifications. Today's lecture builds directly on these ideas.

Key Topics

- Dual function $g(\lambda, \nu)$ and its concavity
- Dual problem as maximization of $g(\lambda, \nu)$ over $\lambda \geq 0$
- Weak duality and duality gap
- Strong duality conditions

Outline

The Lagrangian Dual Problem

Weak Duality

Strong Duality

Economic Interpretation: Shadow Prices

Example: Fractional Knapsack and LP Duality

Proof Space: Key Results

End-of-Lecture Assessment

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Motivation: From Lagrangian to Dual

Recall the constrained optimization problem:

$$\begin{aligned} (P) \quad & \min f(\mathbf{x}) \\ & \text{s.t. } g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m \\ & \quad h_j(\mathbf{x}) = 0, \quad j = 1, \dots, p \end{aligned} \tag{1}$$

The **Lagrangian** function is:

$$L(\mathbf{x}, \lambda, \nu) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^p \nu_j h_j(\mathbf{x})$$

From Lecture 6, KKT conditions tell us that at optimality $\mathbf{x}^*$, there exist λ^*, ν^* such that:

- **Stationarity:** $\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(\mathbf{x}^*) = 0$
- **Dual feasibility:** $\lambda_i^* \geq 0$ for all i
- **Complementary slackness:** $\lambda_i^* g_i(\mathbf{x}^*) = 0$ for all i

Key Insight: Rather than just finding $\mathbf{x}^*$, what if we instead work with the multipliers (λ, ν) as variables?

The Dual Function

For multipliers $\lambda \in \mathbb{R}^m$, $\nu \in \mathbb{R}^p$, define the **dual function**:

$$g(\lambda, \nu) = \inf_{\mathbf{x}} L(\mathbf{x}, \lambda, \nu) = \inf_{\mathbf{x}} \left[f(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^p \nu_j h_j(\mathbf{x}) \right]$$

Key Properties of the Dual Function

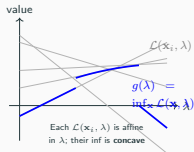
1. **Lower Bound**: For any feasible $\mathbf{x}$ and any $\lambda \geq 0$, $\nu \in \mathbb{R}^p$:

$$g(\lambda, \nu) \leq f(\mathbf{x})$$

This holds because for feasible $\mathbf{x}$: $g_i(\mathbf{x}) \leq 0$, so $\sum \lambda_i g_i(\mathbf{x}) \leq 0$, and $h_j(\mathbf{x}) = 0$.

2. **Concavity**: $g(\lambda, \nu)$ is concave in (λ, ν) **even if f is not convex!**

Why concavity? The infimum of a family of affine functions in (λ, ν) is concave.



The Dual Problem

The **Lagrangian dual problem** is:

$$\begin{aligned} (D) \quad & \max_{\lambda, \nu} \quad g(\lambda, \nu) \\ & \text{s.t.} \quad \lambda \geq 0 \end{aligned} \tag{2}$$

where $g(\lambda, \nu) = \inf_{\mathbf{x}} L(\mathbf{x}, \lambda, \nu)$.

Key Observations

- The dual problem is a **concave maximization problem** (hence convex!)
- The domain of the dual is $\{\lambda \in \mathbb{R}^m : \lambda \geq 0\}$ (no restriction on ν)
- The dual is always convex, **regardless** of whether the primal is convex
- Solving the dual is sometimes easier than solving the primal

Optimal values:

- $p^* = \min_{\mathbf{x}} f(\mathbf{x})$ subject to constraints (primal optimal value)
- $d^* = \max_{\lambda \geq 0, \nu} g(\lambda, \nu)$ (dual optimal value)

Example: LP Dual Derivation

Consider a linear program (standard form):

$$\min \quad \mathbf{c}^T \mathbf{x} \quad \text{s.t.} \quad A\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \geq 0$$

The Lagrangian is:

$$\begin{aligned} L(\mathbf{x}, \lambda, \nu) &= \mathbf{c}^T \mathbf{x} + \sum_{i=1}^m \lambda_i (A_i \cdot \mathbf{x} - b_i) - \sum_{j=1}^n \nu_j x_j \\ &= \sum_{j=1}^n (c_j + \sum_{i=1}^m \lambda_i A_{ij} - \nu_j) x_j - \sum_{i=1}^m \lambda_i b_i \end{aligned}$$

The dual function is:

$$g(\lambda, \nu) = \inf_{\mathbf{x} \geq 0} L(\mathbf{x}, \lambda, \nu) = \begin{cases} -\mathbf{b}^T \lambda & \text{if } \mathbf{c} + A^T \lambda - \nu = 0, \nu \geq 0 \\ -\infty & \text{otherwise} \end{cases}$$

This simplifies to:

$$\max \quad -\mathbf{b}^T \lambda \quad \text{s.t.} \quad A^T \lambda + \nu = \mathbf{c}, \quad \nu \geq 0$$

Or equivalently (dropping ν):

$$\max \quad -\mathbf{b}^T \lambda \quad \text{s.t.} \quad A^T \lambda \leq \mathbf{c}$$

Exercise 1: Computing the Dual Function

Quick Check

Consider the problem: $\min (x - 2)^2$ s.t. $x \geq 1$.

1. Write the Lagrangian $L(x, \lambda)$ where λ is the multiplier for the constraint $1 - x \leq 0$.
2. Compute the dual function $g(\lambda) = \inf_x L(x, \lambda)$ for $\lambda \geq 0$.
3. Verify that $g(\lambda) \leq f(x^*)$ where x^* is the optimal solution.

Solution: Computing the Dual Function

Answers

(1) The constraint is $-x + 1 \leq 0$, so:

$$L(x, \lambda) = (x - 2)^2 + \lambda(1 - x) = (x - 2)^2 - \lambda x + \lambda$$

(2) To find $g(\lambda) = \inf_x L(x, \lambda)$, take the derivative:

$$\frac{\partial L}{\partial x} = 2(x - 2) - \lambda = 0 \implies x^* = 2 + \frac{\lambda}{2}$$

For the constraint $x \geq 1$ to be satisfied: $2 + \frac{\lambda}{2} \geq 1$ always holds for $\lambda \geq 0$.

Thus:

$$g(\lambda) = \left(2 + \frac{\lambda}{2} - 2\right)^2 - \lambda\left(2 + \frac{\lambda}{2}\right) + \lambda = \frac{\lambda^2}{4} - 2\lambda - \frac{\lambda^2}{2} + \lambda = -\frac{\lambda^2}{4} - \lambda$$

(3) The unconstrained minimizer of $(x - 2)^2$ is $x = 2$, which satisfies $x \geq 1$. So the constraint is *not active*, and $x^* = 2$ with $p^* = f(2) = 0$.

Check: $g(\lambda) = -\frac{\lambda^2}{4} - \lambda \leq 0 = p^*$ for all $\lambda \geq 0$. ✓

Note: $\max_{\lambda \geq 0} g(\lambda) = g(0) = 0 = p^*$, so strong duality holds with $\lambda^* = 0$ (consistent with the inactive constraint).

The Lagrangian Dual Problem

Weak Duality

Strong Duality

Economic Interpretation: Shadow Prices

Example: Fractional Knapsack and LP Duality

Proof Space: Key Results

End-of-Lecture Assessment

Weak Duality Theorem

Theorem (Weak Duality)

For any feasible $\mathbf{x}$ in problem (P) and any $\lambda \geq 0, \nu \in \mathbb{R}^p$:

$$g(\lambda, \nu) \leq f(\mathbf{x})$$

In particular, $d^* \leq p^*$ (dual optimal $\leq$ primal optimal).

Proof Sketch

- By definition, $g(\lambda, \nu) = \inf_{\mathbf{x}'} L(\mathbf{x}', \lambda, \nu)$
- For the feasible $\mathbf{x}$: $g_i(\mathbf{x}) \leq 0$ and $h_j(\mathbf{x}) = 0$
- Therefore:

$$g(\lambda, \nu) \leq L(\mathbf{x}, \lambda, \nu) = f(\mathbf{x}) + \sum_i \lambda_i g_i(\mathbf{x}) + \sum_j \nu_j h_j(\mathbf{x}) \leq f(\mathbf{x})$$

since $\lambda_i g_i(\mathbf{x}) \leq 0$ (dual feasible $\lambda_i \geq 0$ and primal feasible $g_i(\mathbf{x}) \leq 0$).

- Taking max over (λ, ν) on left and min over $\mathbf{x}$ on right: $d^* \leq p^*$ ✓

Practical Impact: We can compute $g(\lambda, \nu)$ for any dual feasible (λ, ν) to get a lower bound on the optimal primal value, even without solving the problem!

Duality Gap and Optimality Certificates

Definition (Duality Gap)

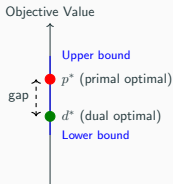
The **duality gap** is defined as:

$$\text{gap} = p^* - d^*$$

Implications of Weak Duality

- Gap ≥ 0 always (by weak duality)
- If we find a feasible x with $f(x) = d^*$, then x is optimal! (gap = 0)
- Even without solving to optimality, if we have dual feasible (λ, ν) with large $g(\lambda, \nu)$, it gives a tight lower bound on p^*
- The dual problem is a *relaxation* of the primal—its optimal value is a lower bound

Geometric Picture:



Exercise 2: Verifying Weak Duality

Quick Check

Consider: $\min x_1^2 + x_2^2$ s.t. $x_1 + x_2 \geq 1$, $(x_1, x_2) \in \mathbb{R}^2$.

1. Rewrite the constraint in standard form $g(x) \leq 0$.
2. For $\lambda = 2$, compute $g(\lambda) = \inf_{\mathbf{x}} L(\mathbf{x}, \lambda)$ where
 $L(\mathbf{x}, \lambda) = \mathbf{x}_1^2 + \mathbf{x}_2^2 + \lambda(1 - \mathbf{x}_1 - \mathbf{x}_2)$.
3. Find the actual primal optimum $\mathbf{x}^*$ (by inspection or calculus).
What is p^* ?
4. Verify that $g(2) \leq p^*$.

Solution: Verifying Weak Duality

Answers

(1) Standard form: $g(\mathbf{x}) = 1 - x_1 - x_2 \leq 0$, i.e., $-x_1 - x_2 + 1 \leq 0$.

(2) $L(\mathbf{x}, \lambda) = x_1^2 + x_2^2 + \lambda(1 - x_1 - x_2)$.

Setting derivatives to zero:

$$\frac{\partial L}{\partial x_1} = 2x_1 - \lambda = 0 \implies x_1 = \frac{\lambda}{2}$$

$$\frac{\partial L}{\partial x_2} = 2x_2 - \lambda = 0 \implies x_2 = \frac{\lambda}{2}$$

Thus:

$$g(\lambda) = \left(\frac{\lambda}{2}\right)^2 + \left(\frac{\lambda}{2}\right)^2 + \lambda\left(1 - \frac{\lambda}{2} - \frac{\lambda}{2}\right) = \frac{\lambda^2}{2} + \lambda(1 - \lambda) = \lambda - \frac{\lambda^2}{2}$$

For $\lambda = 2$: $g(2) = 2 - 2 = 0$.

(3) At the boundary $x_1 + x_2 = 1$ (must be active for optimality),

minimize $x_1^2 + x_2^2$ subject to $x_1 + x_2 = 1$. By symmetry,

$x_1^* = x_2^* = 1/2$, so $p^* = 1/4 + 1/4 = 1/2$.

(4) Check: $g(2) = 0 \leq 1/2 = p^*$. ✓

The Lagrangian Dual Problem

Weak Duality

Strong Duality

Economic Interpretation: Shadow Prices

Example: Fractional Knapsack and LP Duality

Proof Space: Key Results

End-of-Lecture Assessment

Strong Duality: When Gap Becomes Zero

Theorem (Strong Duality for Convex Problems)

If problem (P) is convex and Slater's condition holds (there exists a strictly feasible point), then $d^* = p^*$.

What Strong Duality Means

- The duality gap is zero
- The dual optimal value equals the primal optimal value
- We can solve the easier dual problem to find the primal optimum
- There exist optimal multipliers λ^*, ν^* such that KKT conditions hold

Slater's Condition (Constraint Qualification)

There exists $\tilde{x}$ such that:

- $g_i(\tilde{x}) < 0$ for all $i = 1, \dots, m$ (all inequality constraints strictly satisfied)
- $h_j(\tilde{x}) = 0$ for all $j = 1, \dots, p$ (equality constraints satisfied)

Intuition: If the feasible region has interior points, the duality gap vanishes and optimal multipliers exist.



Strong Duality $\Leftrightarrow$ KKT Conditions

For a convex optimization problem with Slater's condition:

Strong Duality Characterization

$\mathbf{x}^*$ is primal optimal if and only if there exist λ^*, ν^* such that:

1. **Stationarity:** $\nabla f(\mathbf{x}^*) + \sum_i \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_j \nu_j^* \nabla h_j(\mathbf{x}^*) = 0$
2. **Primal feasibility:** $g_i(\mathbf{x}^*) \leq 0, h_j(\mathbf{x}^*) = 0$
3. **Dual feasibility:** $\lambda_i^* \geq 0$
4. **Complementary slackness:** $\lambda_i^* g_i(\mathbf{x}^*) = 0$ for all i

These are precisely the KKT conditions!

Why This Matters

- For convex problems: KKT conditions are **necessary and sufficient** for optimality
- Strong duality says: if KKT conditions hold, $\mathbf{x}^*$ solves both primal and dual
- We can verify optimality without comparing to other feasible points

Exercise 3: Checking Strong Duality Conditions

Quick Check

For each problem, determine if strong duality holds:

1. $\min x_1^2 + x_2^2$ s.t. $x_1 + x_2 = 1$
2. $\min |x|$ s.t. $x \leq 0$ (where $|\cdot|$ is the absolute value)
3. $\min e^x$ s.t. $x^2 \leq 1$
4. $\min -x$ s.t. $x^2 \leq 1$

For each, check: (a) Is the problem convex? (b) Does Slater's condition hold?

Solution: Checking Strong Duality Conditions

Answers

- (1) Yes, strong duality holds. - Convex: objective is convex, equality constraint is affine. - Slater: e.g., $\tilde{x} = (1/2, 1/2)$ is feasible. ✓
- (2) Yes, strong duality holds. - Convex: $f(x) = |x|$ is convex, $g(x) = x \leq 0$ is an affine constraint. - Slater: $\tilde{x} = -1$ satisfies $\tilde{x} < 0$ (strict inequality). ✓
- (3) Yes, strong duality holds. - Convex: e^x is convex, $x^2 - 1 \leq 0$ is convex. - Slater: e.g., $\tilde{x} = 0$ satisfies $0 < 1$. ✓
- (4) Yes, strong duality holds. - Convex: $-x$ is affine (hence convex), $x^2 \leq 1$ is convex. - Slater: e.g., $\tilde{x} = 0$ satisfies $0 < 1$. ✓

Roadmap

The Lagrangian Dual Problem

Weak Duality

Strong Duality

Economic Interpretation: Shadow Prices

Example: Fractional Knapsack and LP Duality

Proof Space: Key Results

End-of-Lecture Assessment

Dual Variables as Shadow Prices

Consider a resource allocation problem:

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{s.t.} \quad g_i(\mathbf{x}) \leq b_i, \quad i = 1, \dots, m$$

Economic Interpretation

The optimal dual variable λ_i^* represents the **shadow price** (or marginal value) of the i -th resource constraint:

$$\lambda_i^* \approx \frac{\partial p^*}{\partial b_i}$$

- If $\lambda_i^* > 0$: increasing the constraint's RHS by 1 unit decreases the optimal cost by λ_i^* (assuming Slater's condition and small perturbations)
- If $\lambda_i^* = 0$: the constraint is not active; perturbing its RHS doesn't help
- Units of λ_i^* : [cost per unit of resource i]

Practical Value

- Dual variables tell us which constraints are "worth relaxing"
- In production planning: λ_i^* is the maximum price to pay for additional resource i
- Guides sensitivity analysis and what-if studies

Sensitivity Analysis Example

Setup: A company manufactures products with time budget and machine constraints.

Primal Problem

$$\min \text{ cost}(\mathbf{x}) \quad \text{s.t.} \quad \text{Time}(\mathbf{x}) \leq 100 \text{ hours}, \quad \text{Machines}(\mathbf{x}) \leq 20 \text{ units}$$

Suppose we solve this and get optimal multipliers:

$$\lambda_{\text{time}}^* = 5 \quad (\$/\text{hour}), \quad \lambda_{\text{machines}}^* = 20 \quad (\$/\text{machine})$$

What This Means

- Buying 1 more hour of production time is worth \$5 in cost reduction
- Buying 1 more machine is worth \$20 in cost reduction
- If we can hire overtime at \$3/hour, it's worthwhile (benefit \$5 > cost \$3)
- If we can rent a machine for \$15/unit, it's worthwhile (benefit \$20 > cost \$15)

Exercise 4: Interpreting Dual Variables

Quick Check

A diet problem: $\min \mathbf{c}^T \mathbf{x}$ (cost of foods) s.t. $A\mathbf{x} \geq \mathbf{b}$ (nutritional requirements), $\mathbf{x} \geq 0$.

The solution yields dual multipliers (shadow prices):

- $\lambda_{\text{protein}}^* = 0.5$ (\$/gram)
- $\lambda_{\text{calcium}}^* = 0$ (\$/gram)
- $\lambda_{\text{vitamin}}^* = 2$ (\$/IU)

Interpret each multiplier. What do they tell us about:

1. Should we relax the protein requirement? By how much?
2. What about calcium?
3. If a multivitamin supplement costs \$1.50 per unit and provides 1 IU, is it worth buying?

Solution: Interpreting Dual Variables

Answers

(1) Protein: $\lambda_{\text{protein}}^* = 0.5 > 0$ means relaxing the protein requirement by 1 gram saves \$0.50 in diet cost. If we can get a cheaper protein source, it's valuable to consider relaxing this constraint. The constraint is *active* (tight).

(2) Calcium: $\lambda_{\text{calcium}}^* = 0$ means the calcium requirement is *not binding*—the optimal diet provides more calcium than required. Relaxing this constraint further provides no cost savings (we're already over-satisfying it).

(3) Vitamin Supplement: The supplement costs \$1.50 but provides only 1 IU, which is worth \$2 in cost reduction (at the current optimum). So it is **worth buying!** It would actually improve the diet.

Note: This assumes we're at an interior solution for vitamin consumption; at the boundary, the analysis changes.

The Lagrangian Dual Problem

Weak Duality

Strong Duality

Economic Interpretation: Shadow Prices

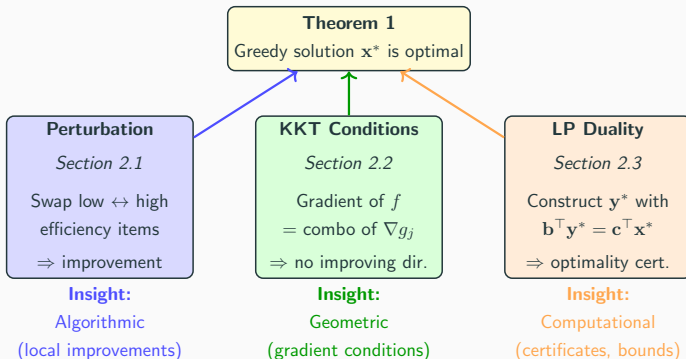
Example: Fractional Knapsack and LP Duality

Proof Space: Key Results

End-of-Lecture Assessment

Overview: Three Proof Strategies for Knapsack

Three Proof Strategies for Theorem 1



Same result, three perspectives $\Rightarrow$ deeper understanding of optimization theory

The Fractional Knapsack Problem

Setup: A merchant has n items with:

- Value c_i for item i
- Weight a_i for item i
- Can take fractional amounts: $0 \leq x_i \leq 1$
- Knapsack capacity: $\sum_{i=1}^n a_i x_i \leq b$ (total weight limit)

Linear Program (Primal)

$$\begin{aligned} (P) \quad & \max_{\mathbf{x}} \quad \mathbf{c}^T \mathbf{x} \\ & \text{s.t.} \quad \mathbf{a}^T \mathbf{x} \leq b \\ & \quad 0 \leq x_i \leq 1, \quad i = 1, \dots, n \end{aligned} \tag{3}$$

Greedy Solution: Sort items by value-to-weight ratio c_i/a_i in descending order. Take items greedily until capacity is reached; take fractional amount of next item if needed.

Key Property: The greedy solution is optimal (for fractional knapsack).
Duality provides a proof!

Fractional vs Binary Knapsack

Fractional vs. Binary Knapsack Solutions

Fractional Knapsack



$$x_1^* = 1 \text{ (full)}$$

$$x_2^* = 1 \text{ (full)}$$

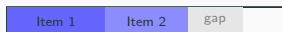
$$x_3^* = 0.67 \text{ (partial)}$$

$$x_4^* = 0$$

$$\text{Optimal: } z_{LP}^* = 28.33$$



Binary (0-1) Knapsack



$$x_1^* = 1 \text{ (take)}$$

$$x_2^* = 1 \text{ (take)}$$

$$x_3^* = 0 \text{ (leave – doesn't fit!)}$$

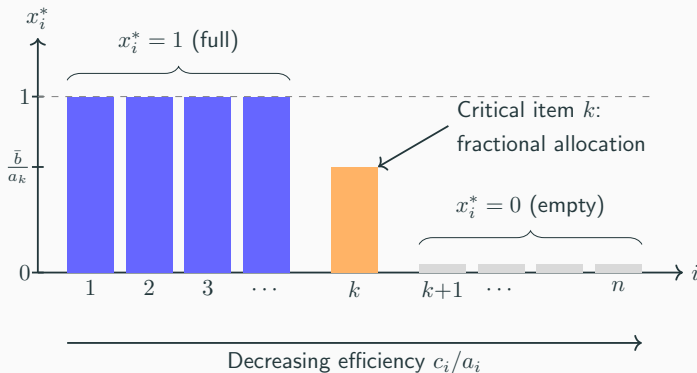
$$x_4^* = 0$$

$$\text{Optimal: } z_{IP}^* = 25$$

$$\text{Integrality gap: } z_{LP}^* - z_{IP}^* = 3.33$$

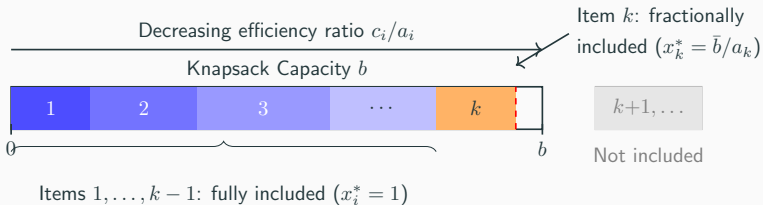
Greedy works for fractional; NP-hard for binary

Greedy Solution Structure



Items are sorted by value-to-weight ratio c_i/a_i in descending order. The optimal solution takes items 1 through k completely, item k fractionally, and nothing after.

Visualizing the Packing Process



Writing the Dual Problem

For the primal:

$$\max \mathbf{c}^T \mathbf{x} \quad \text{s.t.} \quad \mathbf{a}^T \mathbf{x} \leq b, \quad 0 \leq x_i \leq 1$$

The Lagrangian is:

$$\begin{aligned} L(\mathbf{x}, y_0, \mathbf{y}) &= \mathbf{c}^T \mathbf{x} + y_0(b - \mathbf{a}^T \mathbf{x}) + \sum_{i=1}^n y_i(1 - x_i) \\ &= y_0 b + \sum_{i=1}^n (c_i - a_i y_0 - y_i) x_i + \sum_{i=1}^n y_i \end{aligned}$$

The dual function:

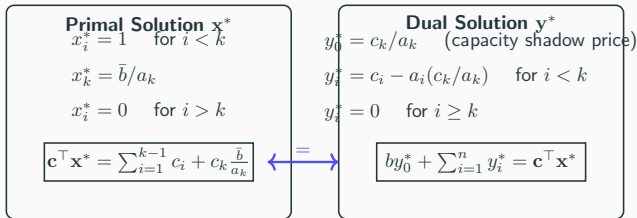
$$g(y_0, \mathbf{y}) = \begin{cases} y_0 b + \sum_i y_i & \text{if } c_i - a_i y_0 - y_i \leq 0 \forall i \\ -\infty & \text{otherwise} \end{cases}$$

Equivalently, constraints are $y_i = \max(0, c_i - a_i y_0)$. The dual problem is:

$$\min_{y_0 \geq 0} \left[b y_0 + \sum_{i=1}^n \max(0, c_i - a_i y_0) \right] \quad (4)$$

Primal-Dual Solution Correspondence

Dual Variables and Economic Interpretation



Shadow Price Interpretation:

$y_0^* = c_k/a_k$: Marginal value of one unit increase in capacity b

$y_i^* > 0$ for $i < k$: Value of relaxing upper bound constraint $x_i \leq 1$

The primal solution structure determines the dual solution: items taken (full or partial) correspond to zero dual variables; items excluded correspond to positive dual multipliers.

Optimal Solution and Shadow Price Interpretation

Greedy Algorithm Analysis:

- Sort by ratio: $\frac{c_1}{a_1} \geq \frac{c_2}{a_2} \geq \dots \geq \frac{c_n}{a_n}$
- Fill items 1, 2, ..., k completely; fill item $k + 1$ fractionally
- Let $y_0^* = \frac{c_k}{a_k}$ (value-to-weight ratio of critical item k)

Optimal Dual Solution

Set:

- $y_0^* = \frac{c_k}{a_k}$ (shadow price of capacity)
- $y_i^* = 0$ for $i \leq k$ (since items 1,..., k are fully taken)
- $y_i^* = a_i y_0^* - c_i > 0$ for $i > k$ (since these items are excluded)

This satisfies $c_i - a_i y_0^* - y_i^* = 0$ for all i ✓

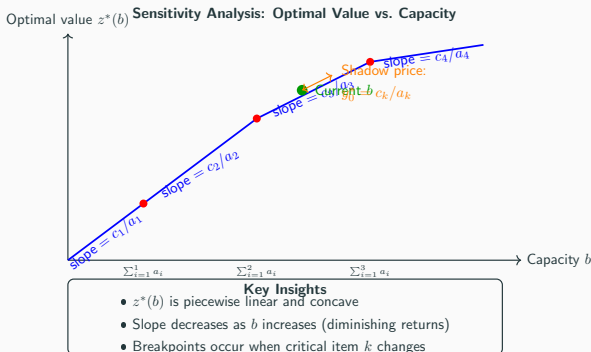
Interpretation

$y_0^* = \frac{c_k}{a_k}$ is the **marginal value of capacity**:

- Adding 1 unit to b increases optimal value by exactly y_0^*
- Items with ratio $> y_0^*$ are included; items with ratio $< y_0^*$ are excluded

Sensitivity Analysis: Optimal Value Function

For knapsack with capacity b , let $z^*(b)$ = optimal value.



Key Observation: The function $z^*(b)$ is **piecewise linear and concave** with slopes equal to value-to-weight ratios. At each capacity, the slope is the shadow price $y_0^*(b)$ (the marginal value of additional capacity).

Exercise 5: Knapsack Duality

Quick Check

A wine merchant has a knapsack with capacity $b = 100$ liters. Available wines:

Wine	Value (\$/liter)	Weight (liters)	Ratio
1	40	5	8
2	30	4	7.5
3	20	3	6.67
4	10	2	5

1. Find the greedy solution (sort by ratio, fill greedily)
2. What is the optimal value p^* ?
3. Construct the optimal dual solution and shadow price y_0^* . Verify optimality by checking that primal and dual objectives are equal.

Solution: Knapsack Duality

Answers

(1) Greedy solution: Sorted by ratio (descending): Wine 1 (ratio 8), Wine 2 (ratio 7.5), Wine 3 (ratio 6.67), Wine 4 (ratio 5).

Fill greedily: total weight = $5 + 4 + 3 + 2 = 14 \ll 100$ (capacity). All items fit! So $\mathbf{x}^* = (1, 1, 1, 1)$ with no fractional item needed.

(2) $p^* = 40 + 30 + 20 + 10 = 100$.

(3) Since all items fit, the capacity constraint is slack, so $y_0^* = 0$. By the dual formulation: $d^* = \min_{y_0 \geq 0} \left[100y_0 + \sum_{i=1}^4 \max(0, c_i - a_i y_0) \right]$. At $y_0^* = 0$: $d^* = 0 + (40 + 30 + 20 + 10) = 100 = p^*$. ✓

The Lagrangian Dual Problem

Weak Duality

Strong Duality

Economic Interpretation: Shadow Prices

Example: Fractional Knapsack and LP Duality

Proof Space: Key Results

End-of-Lecture Assessment

Proof Space: Weak Duality Theorem

Theorem (Weak Duality)

For any feasible $\mathbf{x}$ in (P) and any (λ, ν) with $\lambda \geq 0$:

$$g(\lambda, \nu) \leq f(\mathbf{x})$$

Proof Outline: (to be done on board)

1. Recall definition: $g(\lambda, \nu) = \inf_{\mathbf{y}} [f(\mathbf{y}) + \lambda^T g(\mathbf{y}) + \nu^T h(\mathbf{y})]$
2. For feasible $\mathbf{x}$: $g(\mathbf{x}) \leq 0$ and $h(\mathbf{x}) = 0$
3. Therefore: $f(\mathbf{x}) + \lambda^T g(\mathbf{x}) + \nu^T h(\mathbf{x}) \leq f(\mathbf{x})$ (since $\lambda \geq 0, g(\mathbf{x}) \leq 0, h(\mathbf{x}) = 0$)
4. By definition of infimum:
 $g(\lambda, \nu) \leq f(\mathbf{x}) + \lambda^T g(\mathbf{x}) + \nu^T h(\mathbf{x}) \leq f(\mathbf{x})$
5. Taking minimum over all feasible $\mathbf{x}$: $g(\lambda, \nu) \leq d^* = p^*$

See instructor proof notes for complete details.

Proof Space: Strong Duality under Slater's Condition

Theorem (Strong Duality)

If (P) is convex and Slater's condition holds, then $d^* = p^*$, and there exist optimal multipliers (λ^*, ν^*) .

Proof Strategy: We already know $d^* \leq p^*$ (weak duality). We will show $d^* \geq p^*$ by constructing a dual feasible (λ^*, ν^*) achieving $g(\lambda^*, \nu^*) = f(\mathbf{x}^*)$.

Step 1: Existence of KKT multipliers.

Slater's condition ($\exists \tilde{\mathbf{x}}$ with $g_i(\tilde{\mathbf{x}}) < 0$, $h_j(\tilde{\mathbf{x}}) = 0$) is a *constraint qualification*. It guarantees that if $\mathbf{x}^*$ is primal optimal, then there exist $\lambda^* \geq 0$, $\nu^* \in \mathbb{R}^p$ satisfying the KKT conditions:

1. $\nabla f(\mathbf{x}^*) + \sum_i \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_j \nu_j^* \nabla h_j(\mathbf{x}^*) = 0$ (Stationarity)
2. $g_i(\mathbf{x}^*) \leq 0$, $h_j(\mathbf{x}^*) = 0$ (Primal feasibility)
3. $\lambda_i^* \geq 0$ (Dual feasibility)
4. $\lambda_i^* g_i(\mathbf{x}^*) = 0$ for all i (Complementary slackness)

(The existence of such multipliers under Slater's condition uses a separating hyperplane argument; see Boyd & Vandenberghe §5.2.3 or Bazaraa, Sherali, and Shetty §6.2.)

Proof Space: Strong Duality (continued) — Why Convexity Makes KKT Sufficient

Step 2: Convexity makes $\mathbf{x}^*$ a global minimizer of $\mathcal{L}(\cdot, \lambda^*, \nu^*)$.

Consider $\mathcal{L}(\mathbf{x}, \lambda^*, \nu^*) = f(\mathbf{x}) + \sum_i \lambda_i^* g_i(\mathbf{x}) + \sum_j \nu_j^* h_j(\mathbf{x})$.

Because (P) is convex: f and each g_i are convex, and each h_j is affine.

Since $\lambda_i^* \geq 0$, the Lagrangian $\mathcal{L}(\mathbf{x}, \lambda^*, \nu^*)$ is a **convex function of $\mathbf{x}$** .

The stationarity condition says $\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*) = 0$. For a convex function, **any point with zero gradient is a global minimizer**.

Therefore:

$$\mathbf{x}^* = \arg \min_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda^*, \nu^*) \quad (\text{global, not just local!})$$



This is precisely why convexity is essential—without it, stationarity only gives a critical point.

Step 3: Conclude $g(\lambda^*, \nu^*) = f(\mathbf{x}^*)$.

Since $\mathbf{x}^*$ is a global minimizer of $\mathcal{L}(\cdot, \lambda^*, \nu^*)$:

$$\begin{aligned} g(\lambda^*, \nu^*) &= \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda^*, \nu^*) = \mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*) \\ &= f(\mathbf{x}^*) + \underbrace{\sum_i \lambda_i^* g_i(\mathbf{x}^*)}_{=0 \text{ (comp. slackness)}} + \underbrace{\sum_j \nu_j^* h_j(\mathbf{x}^*)}_{=0 \text{ (primal feas.)}} = f(\mathbf{x}^*) \end{aligned}$$

Step 4: Combine with weak duality.

$$d^* \geq g(\lambda^*, \nu^*) = f(\mathbf{x}^*) = p^* \quad \text{and} \quad d^* \leq p^* \quad \implies \boxed{d^* = p^*}$$

Proof Space: Complementary Slackness

Theorem (Complementary Slackness Condition)

If $\mathbf{x}^*$ is primal optimal, (λ^*, ν^*) are dual optimal, and strong duality holds, then:

$$\lambda_i^* g_i(\mathbf{x}^*) = 0 \quad \text{for all } i = 1, \dots, m$$

Proof. We begin with the weak duality chain and show that strong duality forces every inequality in the chain to be an equality.

Step 1: Establish the inequality chain.

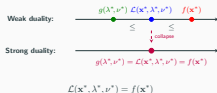
By definition of the dual function (infimum over all $\mathbf{y}$) and primal feasibility of $\mathbf{x}^*$:

$$g(\lambda^*, \nu^*) = \inf_{\mathbf{y}} \mathcal{L}(\mathbf{y}, \lambda^*, \nu^*) \leq \mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*) \leq f(\mathbf{x}^*)$$

The second inequality holds because $\lambda_i^* \geq 0$ and $g_i(\mathbf{x}^*) \leq 0$ imply $\sum_i \lambda_i^* g_i(\mathbf{x}^*) \leq 0$, and primal feasibility gives $\sum_j \nu_j^* h_j(\mathbf{x}^*) = 0$.

Step 2: Strong duality collapses the chain.

Since $f(\mathbf{x}^*) = p^* = d^* = g(\lambda^*, \nu^*)$, the leftmost and rightmost expressions are equal. Therefore **both inequalities are equalities**:



Step 3: Extract the complementary slackness conditions.

Expanding: $f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* g_i(\mathbf{x}^*) + \sum_{j=1}^p \nu_j^* h_j(\mathbf{x}^*) = f(\mathbf{x}^*)$.

Since $h_j(\mathbf{x}^*) = 0$, this reduces to $\sum_{i=1}^m \lambda_i^* g_i(\mathbf{x}^*) = 0$. Each term satisfies $\lambda_i^* g_i(\mathbf{x}^*) \leq 0$ (product of a nonnegative and a nonpositive number). A sum of nonpositive terms equaling zero forces:

$$\lambda_i^* g_i(\mathbf{x}^*) = 0 \quad \text{for each } i = 1, \dots, m$$

Interpretation: For each constraint, either $\lambda_i^* = 0$ (constraint is "free") or $g_i(\mathbf{x}^*) = 0$ (constraint is active/tight). The multiplier is nonzero only for binding constraints.

Proof Space: LP Duality from Lagrangian Duality

Theorem (LP Duality via Lagrangian Duality)

The Lagrangian dual of the primal LP $\min \mathbf{c}^T \mathbf{x}$ s.t. $A\mathbf{x} \geq \mathbf{b}, \mathbf{x} \geq \mathbf{0}$ is the standard LP dual $\max \mathbf{b}^T \mathbf{y}$ s.t. $A^T \mathbf{y} \leq \mathbf{c}, \mathbf{y} \geq \mathbf{0}$.

Proof Outline: (to be done on board)

1. Rewrite primal as: $\min \mathbf{c}^T \mathbf{x}$ s.t. $-A\mathbf{x} + \mathbf{b} \leq \mathbf{0}, -\mathbf{x} \leq \mathbf{0}$
2. Lagrangian: $\mathcal{L}(\mathbf{x}, \lambda, \mu) = \mathbf{c}^T \mathbf{x} - \lambda^T (A\mathbf{x} - \mathbf{b}) - \mu^T \mathbf{x}$ with $\lambda, \mu \geq \mathbf{0}$
3. Rearrange: $\mathcal{L}(\mathbf{x}, \lambda, \mu) = \mathbf{b}^T \lambda + \mathbf{x}^T (\mathbf{c} - A^T \lambda - \mu)$
4. Dual function: $g(\lambda, \mu) = \mathbf{b}^T \lambda + \inf_{\mathbf{x} \geq \mathbf{0}} \mathbf{x}^T (\mathbf{c} - A^T \lambda - \mu)$
5. Infimum is 0 if $\mathbf{c} - A^T \lambda - \mu \geq \mathbf{0}$ (i.e., $A^T \lambda + \mu \leq \mathbf{c}$), else $-\infty$
6. Set $\mathbf{y} = \lambda$ and optimize: $\max_{\lambda, \mu} \mathbf{b}^T \lambda$ s.t.
 $A^T \lambda + \mu = \mathbf{c}, \lambda \geq \mathbf{0}, \mu \geq \mathbf{0}$
7. This simplifies to $\max \mathbf{b}^T \mathbf{y}$ s.t. $A^T \mathbf{y} \leq \mathbf{c}, \mathbf{y} \geq \mathbf{0}$ (standard LP dual)

See instructor proof notes for complete details.

Exercise: Building a Dual from Scratch

Group Exercise (10 min)

Consider the problem:

$$\min (x_1 - 2)^2 + (x_2 - 2)^2 \quad \text{s.t.} \quad x_1 + x_2 \leq 2$$

Suppose you already know: $\mathbf{x}^* = (1, 1)$, $\lambda^* = 2$, $p^* = f(\mathbf{x}^*) = 2$.

1. Write the Lagrangian $\mathcal{L}(x_1, x_2, \lambda)$ for $\lambda \geq 0$.
2. Minimize $\mathcal{L}$ over $\mathbf{x} \in \mathbb{R}^2$ (unconstrained). Set $\nabla_{\mathbf{x}} \mathcal{L} = 0$ and solve for $x_1(\lambda)$, $x_2(\lambda)$.
3. Substitute back to get the dual function $g(\lambda) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda)$. Simplify.
4. Solve the dual: $\max_{\lambda \geq 0} g(\lambda)$. Find λ_{dual}^* and $d^* = g(\lambda_{\text{dual}}^*)$.
5. Compare d^* vs. p^* . Compare λ_{dual}^* with the KKT multiplier.

Hint: By symmetry, $x_1(\lambda) = x_2(\lambda)$. The dual function will be a concave quadratic.

Solution: Building a Dual from Scratch

Step-by-Step Derivation

(1) $\mathcal{L}(x_1, x_2, \lambda) = (x_1 - 2)^2 + (x_2 - 2)^2 + \lambda(x_1 + x_2 - 2), \quad \lambda \geq 0.$

(2) $\frac{\partial \mathcal{L}}{\partial x_1} = 2(x_1 - 2) + \lambda = 0 \Rightarrow x_1(\lambda) = 2 - \frac{\lambda}{2}.$ By symmetry,
 $x_2(\lambda) = 2 - \frac{\lambda}{2}.$

(3) Substituting: $g(\lambda) = 2 \left(\frac{\lambda}{2}\right)^2 + \lambda \left(2(2 - \frac{\lambda}{2}) - 2\right) = \frac{\lambda^2}{2} + 2\lambda - \lambda^2$

$$g(\lambda) = -\frac{\lambda^2}{2} + 2\lambda$$

(4) $g'(\lambda) = -\lambda + 2 = 0 \Rightarrow \lambda_{\text{dual}}^* = 2. \quad g(2) = -2 + 4 = 2.$
 $d^* = 2.$

(5) $d^* = 2 = p^*$: Strong duality! $\lambda_{\text{dual}}^* = 2 = \lambda_{\text{KKT}}^*$: They're the same!

Verify weak duality: $g(\lambda) = -\frac{1}{2}(\lambda - 2)^2 + 2 \leq 2 = p^*$ for all $\lambda \geq 0$. ✓

The KKT–Duality Connection

Key insight: At a KKT point $(\mathbf{x}^*, \lambda^*)$ for a convex problem:

1. **Stationarity** $\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \lambda^*) = 0$

$\Rightarrow \mathbf{x}^*$ minimizes $\mathcal{L}(\cdot, \lambda^*)$ (since $\mathcal{L}$ is convex in $\mathbf{x}$)

$\Rightarrow g(\lambda^*) = \mathcal{L}(\mathbf{x}^*, \lambda^*)$

2. **Complementary slackness** $\lambda_i^* g_i(\mathbf{x}^*) = 0$

$$\Rightarrow \mathcal{L}(\mathbf{x}^*, \lambda^*) = f(\mathbf{x}^*) + \underbrace{\sum_i \lambda_i^* g_i(\mathbf{x}^*)}_{=0} = f(\mathbf{x}^*) = p^*$$

3. **Combining:** $g(\lambda^*) = p^* \Rightarrow$ **zero duality gap**, and λ^* solves the dual

KKT multipliers = dual optimizers = shadow prices.

They are the same object seen from three viewpoints.

The unifying condition: For convex problems,

$$\text{Slater's CQ} \implies \text{MFCQ} \implies \text{KKT holds} \iff \text{strong duality holds}$$

The **same constraint qualification** that makes KKT necessary also closes the duality gap.

Exercise: Slater $\Rightarrow$ MFCQ

Proof Exercise

Consider a convex problem where all g_i are convex and differentiable. Suppose Slater's CQ holds: $\exists \hat{\mathbf{x}}$ with $g_i(\hat{\mathbf{x}}) < 0$ for all i .

Let $\bar{\mathbf{x}}$ be any feasible point with active set $\mathcal{A}(\bar{\mathbf{x}}) = \{i : g_i(\bar{\mathbf{x}}) = 0\}$.

1. Using convexity of g_i , show that $\nabla g_i(\bar{\mathbf{x}})^T (\hat{\mathbf{x}} - \bar{\mathbf{x}}) \leq g_i(\hat{\mathbf{x}}) - g_i(\bar{\mathbf{x}})$.
2. For $i \in \mathcal{A}(\bar{\mathbf{x}})$, deduce that $\nabla g_i(\bar{\mathbf{x}})^T (\hat{\mathbf{x}} - \bar{\mathbf{x}}) < 0$.
3. Conclude that $d = \hat{\mathbf{x}} - \bar{\mathbf{x}}$ satisfies the MFCQ condition at $\bar{\mathbf{x}}$.

Hint: For active constraints, $g_i(\bar{\mathbf{x}}) = 0$ and $g_i(\hat{\mathbf{x}}) < 0$.

Solution: Slater $\Rightarrow$ MFCQ

Proof

(1) By the first-order characterization of convexity:

$$g_i(\hat{\mathbf{x}}) \geq g_i(\bar{\mathbf{x}}) + \nabla g_i(\bar{\mathbf{x}})^T (\hat{\mathbf{x}} - \bar{\mathbf{x}})$$

Rearranging: $\nabla g_i(\bar{\mathbf{x}})^T (\hat{\mathbf{x}} - \bar{\mathbf{x}}) \leq g_i(\hat{\mathbf{x}}) - g_i(\bar{\mathbf{x}})$.

(2) For $i \in \mathcal{A}(\bar{\mathbf{x}})$: $g_i(\bar{\mathbf{x}}) = 0$ (active) and $g_i(\hat{\mathbf{x}}) < 0$ (Slater). So:

$$\nabla g_i(\bar{\mathbf{x}})^T (\hat{\mathbf{x}} - \bar{\mathbf{x}}) \leq g_i(\hat{\mathbf{x}}) - 0 = g_i(\hat{\mathbf{x}}) < 0$$

(3) Setting $d = \hat{\mathbf{x}} - \bar{\mathbf{x}}$: we have $\nabla g_i(\bar{\mathbf{x}})^T d < 0$ for all $i \in \mathcal{A}(\bar{\mathbf{x}})$. This is exactly the MFCQ condition!

Key insight: The Slater point $\hat{\mathbf{x}}$ provides a natural MFCQ direction at every feasible point. The direction from $\bar{\mathbf{x}}$ toward $\hat{\mathbf{x}}$ moves strictly into the interior of every active constraint. This is why Slater's condition is so powerful — it guarantees MFCQ globally, not just at a single point.

Exercise: MFCQ and the KKT Multiplier Set

Challenge Exercise

Let $\bar{\mathbf{x}}$ be a KKT point. Define the multiplier set:

$$\Lambda(\bar{\mathbf{x}}) = \left\{ (\lambda_1, \dots, \lambda_m) \geq 0 \mid \nabla f(\bar{\mathbf{x}}) + \sum_i \lambda_i \nabla g_i(\bar{\mathbf{x}}) = 0, \lambda_i g_i(\bar{\mathbf{x}}) = 0 \right\}$$

1. Show that $\Lambda(\bar{\mathbf{x}})$ is a convex set.
2. Show that if MFCQ holds at $\bar{\mathbf{x}}$, then $\Lambda(\bar{\mathbf{x}})$ is bounded.

Hint: Suppose not. Take $\lambda^k \in \Lambda(\bar{\mathbf{x}})$ with $\|\lambda^k\| \rightarrow \infty$. Normalize to get $\bar{\lambda}^k = \lambda^k / \|\lambda^k\|$, extract a limit $\bar{\lambda} \neq 0$ with $\sum \bar{\lambda}_i \nabla g_i(\bar{\mathbf{x}}) = 0$, $\bar{\lambda} \geq 0$. Take inner product with the MFCQ direction d .

3. Relate to the examples: in the exercise where LICQ held, $|\Lambda| = 1$. In the exercise where only MFCQ held, Λ was a line segment. What happens when MFCQ fails?

Solution: MFCQ and the KKT Multiplier Set

Answers

(1) If $\lambda, \lambda' \in \Lambda(\bar{\mathbf{x}})$ and $\alpha \in [0, 1]$, then $\alpha\lambda + (1 - \alpha)\lambda'$ satisfies stationarity (linear combination of two zero vectors is zero), complementary slackness (each term $(\alpha\lambda_i + (1 - \alpha)\lambda'_i)g_i = 0$), and $\alpha\lambda + (1 - \alpha)\lambda' \geq 0$. So $\Lambda(\bar{\mathbf{x}})$ is convex.

(2) Suppose $\Lambda(\bar{\mathbf{x}})$ is unbounded. Take $\lambda^k \in \Lambda$ with $r_k = \|\lambda^k\| \rightarrow \infty$. Let $\bar{\lambda}^k = \lambda^k / r_k$ (unit sphere). Extract convergent subsequence: $\bar{\lambda}^k \rightarrow \bar{\lambda}$ with $\|\bar{\lambda}\| = 1$, $\bar{\lambda} \geq 0$.

Dividing stationarity by r_k : $\nabla f / r_k + \sum \bar{\lambda}_i^k \nabla g_i = 0$. Taking $k \rightarrow \infty$:

$$\sum_{i \in \mathcal{A}} \bar{\lambda}_i \nabla g_i(\bar{\mathbf{x}}) = 0, \quad \bar{\lambda} \geq 0, \quad \bar{\lambda} \neq 0$$

Inner product with MFCQ direction d : $\sum \bar{\lambda}_i \nabla g_i^T d = 0$. But each term is ≤ 0 with at least one < 0 . **Contradiction!**

(3) LICQ: Λ is a single point. MFCQ: Λ is bounded (compact). No MFCQ: Λ is unbounded or empty.

Roadmap

The Lagrangian Dual Problem

Weak Duality

Strong Duality

Economic Interpretation: Shadow Prices

Example: Fractional Knapsack and LP Duality

Proof Space: Key Results

End-of-Lecture Assessment

Exercise 6: Assessment

Assessment: True/False and Short Answer

1. **True or False:** The dual function $g(\lambda, \nu)$ is always convex, even if the primal problem is non-convex.
2. **True or False:** Weak duality implies that $p^* \geq d^*$.
3. **Short Answer:** State Slater's condition for the problem: $\min f(\mathbf{x})$
s.t. $g_i(\mathbf{x}) \leq 0$, $h_j(\mathbf{x}) = 0$.
4. **Short Answer:** In the knapsack problem, what is the economic meaning of the optimal dual variable y_0^* ?
5. **True or False:** If the duality gap is zero, then the KKT conditions must hold (assuming convexity and Slater's condition).

Answers

(1) **False.** The dual function $g(\lambda, \nu) = \inf_{\mathbf{x}} L(\mathbf{x}, \lambda, \nu)$ is the infimum of affine functions in (λ, ν) , hence **concave**, not convex. (The dual problem—maximizing a concave function—is a convex optimization problem, but g itself is concave.) This holds regardless of the convexity of f .

(2) **True.** Weak duality states $d^* \leq p^*$, which is equivalent to $p^* \geq d^*$.

(3) **Slater's Condition:** There exists $\tilde{\mathbf{x}}$ such that $g_i(\tilde{\mathbf{x}}) < 0$ for all i (strict inequality for inequality constraints) and $h_j(\tilde{\mathbf{x}}) = 0$ for all j (exact equality for equality constraints).

(4) y_0^* is the **shadow price** of the capacity constraint: it represents the marginal increase in optimal value per unit increase in capacity. Economically, it's the maximum price one should pay for additional capacity.

(5) **True.** For a convex problem with strong duality and zero gap, there exist optimal dual variables satisfying the KKT conditions. (If $d^* = p^*$ with Slater's and convexity, KKT conditions hold at any primal-dual optimal pair.)

Key Takeaways

Main Concepts

- **Dual Function:** $g(\lambda, \nu) = \inf_{\mathbf{x}} L(\mathbf{x}, \lambda, \nu)$ is concave, providing lower bounds on primal optimal value
- **Weak Duality:** $d^* \leq p^*$ always holds; measures distance from optimality
- **Strong Duality:** For convex problems satisfying Slater's condition, $d^* = p^*$; enables certificate of optimality
- **Dual Variables:** Optimal λ^* and ν^* have economic interpretation as shadow prices/sensitivity coefficients
- **Connection to KKT:** Strong duality (with convexity) $\Leftrightarrow$ KKT conditions provide necessary and sufficient optimality

Practical Implications

- Duality enables optimization without explicitly solving the primal
- Dual variables guide sensitivity analysis (how much to relax which constraints?)
- Many algorithms (cutting planes, column generation, subgradient methods) exploit duality
- Fractional knapsack: duality provides proof of greedy algorithm optimality

Next Time: Lecture 8 - Gradient Descent

Lecture 8 Preview

- **Problem:** How do we *compute* optimal solutions efficiently?
- **Approach:** Start with simplest method: gradient descent
 - Update rule: $\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \nabla f(\mathbf{x}_k)$
 - Move downhill along negative gradient; larger step α_k when gradient is small
- **Analysis:** Convergence guarantees
 - How fast do we converge?
 - How does convexity/smoothness/strong convexity affect convergence rate?
- **Connection to Duality:** Gradient descent relates to Lagrangian dual ascent methods

Big Picture: Duality is the *theory*; algorithms are the *practice*.

Understanding duality helps us design better algorithms.

Questions?

Duality Theory – Key Ideas:

- The *dual problem* complements the primal problem
- *Weak duality* always holds; provides bounds without solving
- *Strong duality* (under Slater's condition) connects duality to KKT and optimality
- *Shadow prices* (dual variables) have economic meaning

Next class: Bring questions about gradient descent, convergence rates, and algorithm analysis.

Office hours: See syllabus for times. Feel free to ask about duality intuition, applications, or connections to your research!